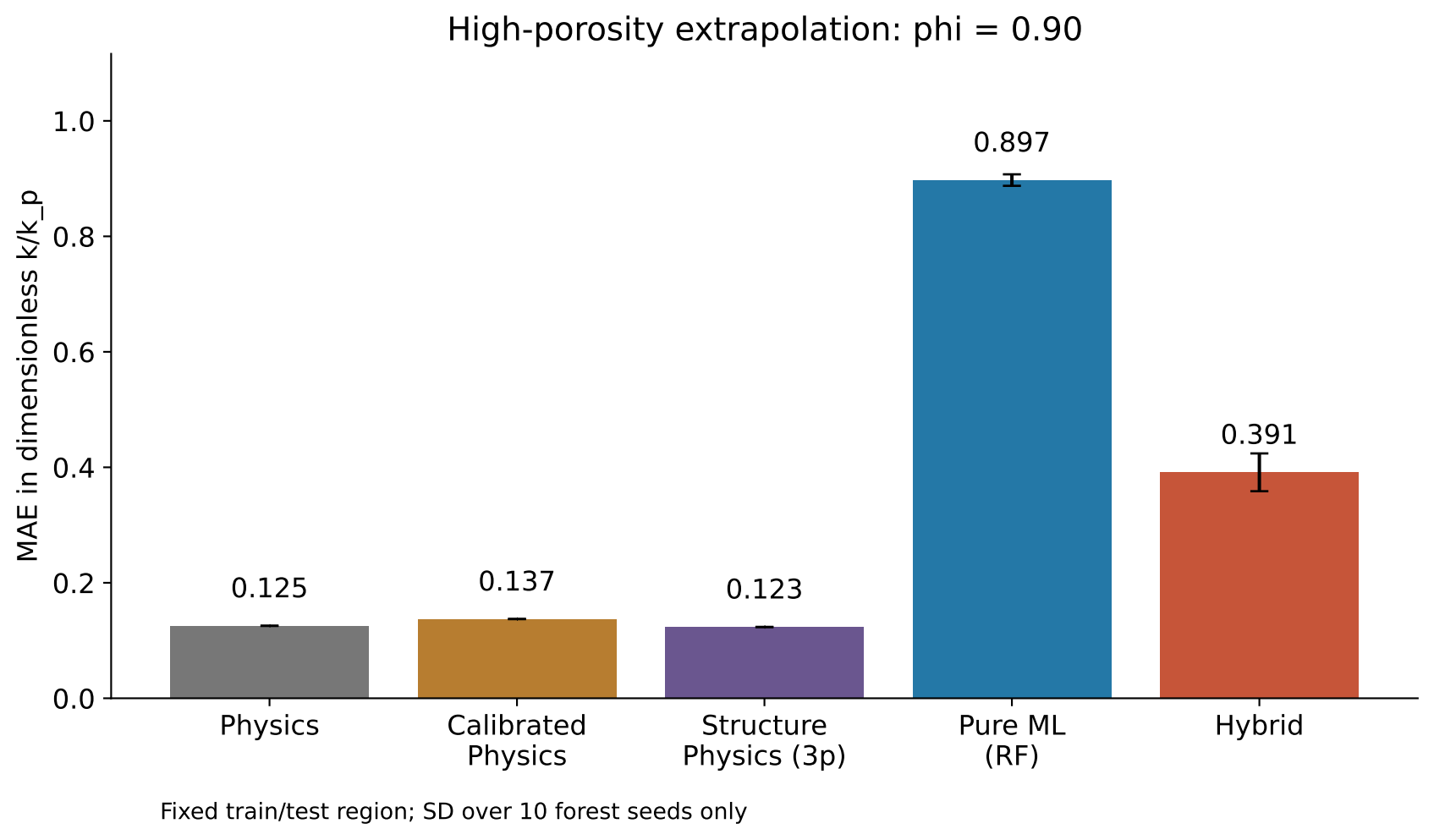
High-porosity extrapolation: phi = 0.90
MAE in dimensionless k/k_p
1.0
0.8
0.6
0.4
0.2
0.0
0.125
0.137
0.123
0.897
0.391
Physics
Calibrated Physics
Structure Physics (3p)
Pure ML (RF)
Hybrid
Fixed train/test region; SD over 10 forest seeds only